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Emotion Detection of Textual Data: An Interdisciplinary Survey

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Abstract—Emotion is a primary aspect of communication and can be expressed in many modalities. Text-Based Emotion Detection (TBED), one of the fastest growing branches of Natural Language Processing (NLP), is the process of classifying syntactic or semantic units of a corpus into a given set of emotion classes proposed by a psychological model. Automatic TBED mechanisms use machine learning approaches to create computational platforms automating the process of extracting emotions. TBED has a wide variety of applications in the area of artificial intelligence: Semantic analysis of documents and public messages related to terrorist attacks (to mitigate risks), automated analysis of historical corpora, and study of product reviews (to assess customer satisfaction). This work reviews the current literature of TBED and the psychological models associated with them.

Index Terms—Text-based emotion detection, psychological model, emotion lexicon, emotion dataset.

I. INTRODUCTION

Natural Language Processing (NLP) is the branch of artificial intelligence that involves automating the preparation and analysis of large amounts of textual and verbal data. One of the main focuses of NLP is to make use of machine learning approaches [1]–[4] to design and construct computational platforms. These platforms automate the process of extracting knowledge from both structured and unstructured sources to facilitate searching through large textual data in a short period of time. Transfer learning models [5]–[9] and Sentiment analysis [10] can be used to enhance emotion detection techniques. These techniques can improve models in artificial intelligence and robotics applications [11]–[13]. Also using domain adaptation techniques in transfer learning models can improve emotion analysis of textual data [14], [15]. Text-Based Emotion Detection (TBED) has been used by researchers to automatically detect affect, identifying the feelings and sentiments expressed in a text. It utilizes methods of computational linguistics, text analysis, machine learning, and NLP to detect emotions from various data sources.

The paper is organized as follows: Section II provides an extensive review on theories of emotions. Then, Section III investigates prior work based on the detection

of text-based emotion. Next, we describe emotion-related data sets in Section IV, and Section V reviews emotion-related lexicons. Finally, we identify open issues in the field and areas for future work in Section VI, and summarize our contributions in Section VII.

II. THEORIES OF EMOTIONS

Emotion is a construct that describes the feelings of an individual or group [16]. Studies to categorize emotions date back to the ancient Greeks and Romans. In the 19th century, Charles Darwin and William James further described evolutionary theories of emotions [17], [18]. Then in 1991, Turkle expressed the notion that a computer is not just a mathematical object, but instead, that they can be used to interpret emotions, languages, and cultures [19]. In the early nineties, Scherer also designed an expert system that sought to ascribe an emotional label as the output of emotional experience information [20]. Furthermore, in 2003, Picard inspected emotional mechanisms in human life and presented a view of affective computation [21]. Picard proposed that emotional intelligence is missing in computing, and encouraged developers to consider imbuing them with emotion recognition capabilities to assist in decision making. Then in 2005, Boehner et al. critiqued measuring emotions as discrete elements, and offered an alternative-dynamic interface of emotions [22].

A. Ekman

Previously Ekman defined a model of emotions, and described the basic constituents as being: anger, disgust, fear, happiness, sadness, and surprise. Ekman's model is very useful for the application of facial emotion detection. However, there is doubt that Ekman's model completely represents the full spectrum of emotions, and that it is limited in that it solely considers the emotions of Western cultures [23].

B. Parrot

Parrot's model, as shown in Fig. 1, also considers six basic emotions, consisting of fear, sadness, surprise,

anger, love, and joy [24]. However, Parrot extended these larger categorizations and arranged them in a tree, that ultimately encompasses 100 separate emotions.

C. A Circumplex Model

As a set of independent and correlated affective dimensions, Russell introduced a circumplex model as shown in Fig. 2a in two dimensions along two orthogonal axes which plots 150 affective labels. Arousal (activation) in the vertical axis represents deactivation in negative, neutral in center, and activation in the positive side, whereas the horizontal axis performs unpleasant in the negative side, medium in center, and pleasant in the positive side. This circumplex model includes pleasure, excitement, arousal, distress, displeasure, depression, sleepiness, and relaxation [25], [26]. Russell et al. described the affective dimensions (anger, fear, shame, jealousy, etc.) as horizontally as part of circumplex and vertically as a fuzzy hierarchy. They believed emotion can be divided into comprehensible entities [27]. In this model, similar emotions like frustrated, distressed, and annoyed are grouped close together and dissimilar emotions are placed further apart. Unlike Ekman's model, Russell's model points to relationships between emotions. For example, with feeling depressed there is an expectation to experience little to no feelings of happiness [23].

D. Scherer's Update to the Russell's Model

Russell's model was used to classify emotions exactly on an edge of circumplex shape, which for every x and y coordinates applies the equation of $X^2 + Y^2 = R^2$. However, it does not cover the emotions in the circle. To remedy this concern, Scherer created a model that for every x and y emotions there exists equation $(x_h)^2 + (y_k)^2 = R^2$, which covers the whole circle. In this way, the model covers the emotions, for example, zero valence and small negative level of arousal. This model is bi-dimensional of arousal and valence and illustrates a broad range of emotions [23], [28]. Georgies et al. also used Scherer's model for classifying blog posts in two dimensions of emotion analysis to predict the level of valence and arousal of each text [29].

E. Whissell

Whissell suggested a continuous bi-dimensional emotion model as illustrated in Fig. 2b whose dimensions are evaluation and activation in a pair of <activation, evaluation> which are assigned to the words from the Dictionary of Affect (approximately 9000 words). The evaluation dimension is about pinpointing feeling from negative to positive, and the activation dimension is measured by taking action from active to passive [25].

F. Plutchik

Plutchik proposed a three-dimensional hybrid model as shown in Fig. 3a which consists of eight basic-complex emotions (joy, sadness, anger, fear, trust, disgust, surprise, and anticipation). He arranged emotions like a color wheel in four groups of primary, secondary, tertiary, and opposite emotions. In the emotion wheel, basic emotions are located in the inner portion and become more complex in the outer portions. Each of them is further sub-divided into three ranges of emotions, where those close to the inner circle are more intense. For example, anticipation is more complex than vigilance but less intense. Interest is more complex than anticipation and less intense. Also, similar emotions are organized closely together, and opposite emotions are 180 degrees apart from each other [30].

G. OCC Model

Ortony et al. offered an OCC model consisting of 22 emotion categories in three sources: goals, standards, and tastes, and each of them are the basis of three types of events, actions, and objects. Hence, events describe desirability or undesirability of goals, actions explain praise or blameworthiness of standards, and objects reflect appealing or unappealing tastes of individuals. Moreover, some emotions are caused by mixing two or three types of emotions [16], [34]. Steunebrink et al. critiqued the OCC model, and offered a computer scientist's perspective, suggesting ways to make it more useful, practical, and computable.

H. Hourglass of Emotions

The Hourglass model of emotions is shown in Fig. 3b, and consists of 20 categories (half positive and half negative) in four independent dimensions [33]. It is based off of Plutchick's model. Each emotion is presented in a pair of words in order to indicate the root of the word. Cambria et al. claimed that their proposed model can explain the entire emotional experience that happens to everyone. As an example, the difference between guilt and shame is the negativity between the self and an act. Guilt is caused by believing a bad thing was done. Shame is caused when an individual thinks they are a bad person. The difference is very important because it displays distinct results and satisfactorily represents similarities and differences of the effective words.

I. Fontaine

Fontaine et al. studied three languages and found that the four-dimension emotional model can present the full range of emotional experiences in everyone. The four spaces that they applied on the three languages are evaluation- pleasantness, potency- control, activation- arousal, and unpredictability. They studied six major

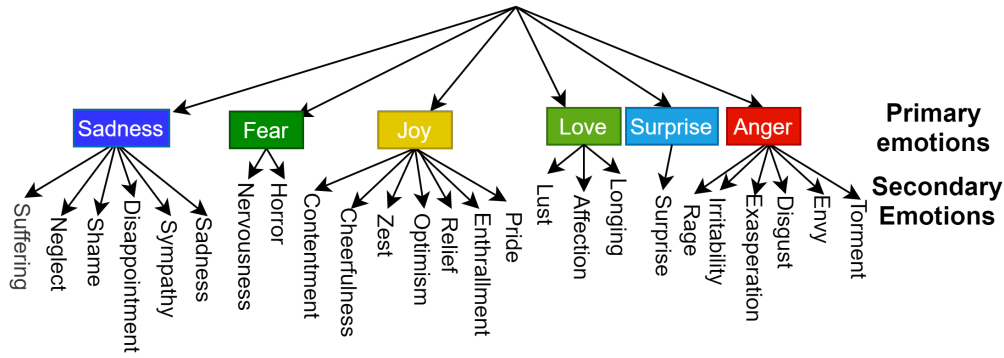


Fig. 1: Parrot's emotions model

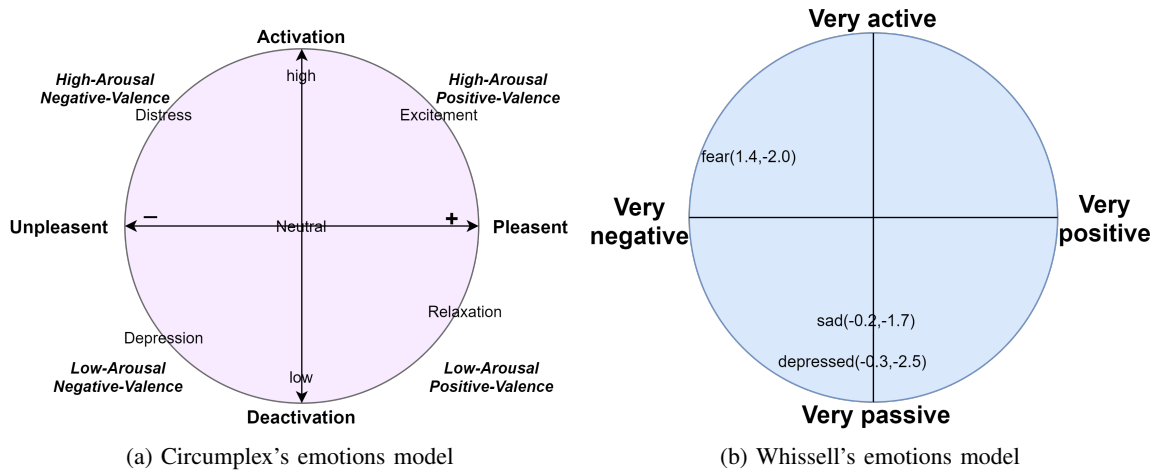


Fig. 2: Examples of 2D emotion models, Circumplex (a) and Whissell models (b)

emotion components that, in total, comprise 144 features [35].

III. TEXT-BASED EMOTION DETECTION

TBED mechanisms can be generally categorized into three classes: rule-based, statistic-based, and hybrid.

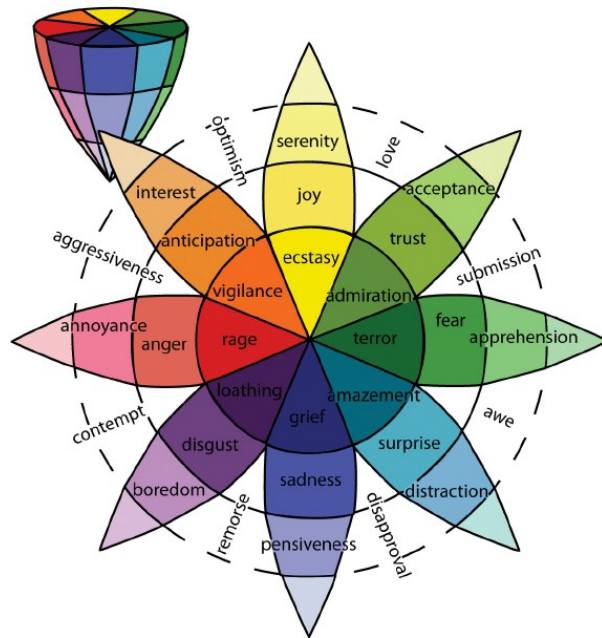
Cherry et al. presented two learning models for TBED in suicide notes. They got fourth place in a competition with 55.22% F_1 -score. Their latent sequence model is a multi-label sentence classifier which annotates with zero or more emotions. Their system uses one classifier per emotion, and it simplifies label balance and fast development issues. It is a binary classifier and produces a stronger result. They used fifteen emotions and Roget's thesaurus to use synonyms of those emotions [36].

Mohammad et al. created a term-emotion lexicon manual annotation through Amazon's Mechanical Turk service and called it NRC (Emolex), which included intensity quantifiers. Their proposed lexicon is larger than WAL (WordNet-Affect) on Plutchik model [37].

Bandhakavi et al. created UMM (unigram mixture model) for TBED. They claimed the lexicon features

outperform state-of-the-art performance using a combination of n-grams, part-of-speech information and sentiment lexicons. To evaluate their model, they used a set of four data sets: Twitter data set and emotion labels by Parrot's primary emotions; Incident reports data set (ISEAR) consisting of 7,000 incident reports with seven basic emotions; News data set (SemEval-2007) consisting of 1250 emotional news headlines using Ekman's basic emotions and annotated with emotion ratings in interval [-100, 100]; and finally Blog dataset which consists of 5,500 blog sentences annotated by three annotators with Ekman's basic emotions and the average kappa agreement of 0.76. The best performing result was on blog data with 0.60 F_1 . In this paper they studied emotion feature extraction using DSELs (domain specific lexicons) and GPLeLs (general purpose emotion lexicons) [38].

Aman et al. applied SVM on blogs for classifying emotions into seven classes. The features were corpus-based unigram and comprise words in emotion lexicons extracted from Roget's thesaurus and WNA. They combined these features and found good results in machine



(a) Plutchik's emotions wheel, [31]. Figure taken from [32].



(b) Hourglass's emotions model, and Figure taken from [33].

Fig. 3: Examples of dimensional emotion models (Plutchik and Hourglass models)

learning experiments. Mixed-emotion label was used when either more than one or no emotion fits on 21 observations. F_1 ranged 0.493 to 0.751 for each class [39].

Kozareva et al. studied emotion classification based on frequency of words collected from MyWay, AlltheWeb, and Yahoo. They claimed words that have a high frequency through many texts with a given emotion have a possibility to express the emotion. They used six emotions for this study which are anger, fear, sadness, surprise, disgust, and joy. They combined all the frequency word counts of the three resources and measured MIs (Mutual Information Scores) of a bag of content and emotion words.

Supervised learning techniques are time-consuming and expensive due to using more than two annotators. Kim et al. presented unsupervised learning techniques for automatically detecting emotions in text. They proposed two models by using four emotions (anger, fear, joy, and sadness), which are common in SemEval-2007, ISEAR and children's fairy tales data sets. Their experiment with two models and three data sets shows that a categorical model using Non-negative Matrix Factorization results in a better performance for SemEval and fairy tales, whereas a dimensional model performs better with ISEAR. The highest performance of F_1 of 0.733 for Non-negative matrix factorization among the three models was reported [40].

Wang et al. collected one of the largest data sets from Twitter containing about 2.5 million tweets for automatic TBED. They applied two machine learning classifiers (LIBLINEAR and MNB) after data preprocessing with seven emotion categories for emotion identification. Also, they used LIWC dictionary and MPQA lexicon for polarities analysis with the feature combination of unigrams, bigrams, existing sentiment, and part of speech. They achieved the maximum accuracy of 65.57% for this study [41].

Tokuhsa et al. automatically generated a collection of 1.3 million emotional events from the web in Japanese as an emotion lexicon. For building an EP corpus, they created a handcrafted lexicon of emotion words classified into ten emotions from a Japanese dictionary, and identified 349 emotion words. Next, they applied the emotion lexicon to the Japanese web corpus, which contains 500 million sentences, and collected 1.3 million events as emotion causes. Then, they evaluated 2000 randomly chosen events by an annotator. Next, they proceeded with a two-step method for emotion classification task consisting of sentiment polarity classification and emotion classification [42].

Strapparava et al. automatically analyzed emotions in text. They used a large data set from News headlines, extracted from news websites (such as Google news and

Work	year	Corpus	Lexicon	no. of labels	Method	P	R	F_1
[45]	2021	Tweets	-	4	BERT	-	-	0.89
[44]	2020	Fairy Tales	AWNA	4	NMF	-	-	0.81
[40]	2010	Fairy tales	WNA	4	NMF	0.75	0.73	0.73
[39]	2008	Blog	-	6	Unigram	0.77	0.45	0.57
[36]	2012	Suicide notes	-	15	SVM LVM	-	-	0.55
[43]	2008	Headlines	-	6	LSA	0.10	0.90	0.17
[48]	2015	Sina	-	8	eLDM	0.64	0.63	0.64
[49]	2013	Tweets	Olympex PMI-Hash	20	Dystemo	0.37	0.44	0.41
[50]	2012	Tweets	TEC	6	ngrams	0.55	0.46	0.50
[38]	2017	Tweets	-	6	UMM	-	-	0.64

TABLE I: Comparison of TBED mechanisms based on their precision, recall, and F_1 .

CNN) and/or newspapers and annotated for Ekman's emotions. The two data sets are a train data set consisting of 250 annotated headlines, and a test data set consisting of 1,000 annotated headlines. They proposed five systems for the automatic identification of those emotions in text, which are WNA presence, LSA single word, LSA emotion synset, LSA all emotion words, and NB trained on blogs. WNA presence provides the best precision (38.28%), and the LSA all emotion words leads to the highest recall of 90.22% and F_1 of 17.57% [43].

Zad et al. applied Non-negative Matrix Factorization (NMF), Principle Component Analysis (PCA), and Latent Dirchelet Allocation (LDA) on Alm's Fairy tales data set, and respectively the overall F_1 are reported 80.9%, 76%, and 60.1%. They applied dimension reduction methods on 1,090 sentences of Alm's fairy tales that all agreed on labels by two annotators from two categories of emotion and mood. They applied WNA lexicon and augmented it for four emotions [44].

Chiorrini et al. applied deep learning and evaluated Bidirectional Encoder Representations from Transformers (BERT) model on Twitter data, and reported 89% F_1 for emotion detection on four emotions [45]. Then Akhtar et al. developed three deep learning models based on CNN, LSTM, and GRU and one supervised model based on SVR. The pre-trained word embedding models of GloVe and word2vec were trained. The proposed model was evaluated on the benchmark setup of EmoInt-2017 and SemEval-2017. They achieved the best accuracy of 74.8% on generic tweets for emotion intensity prediction [46]. Next, Krommyda et al. applied LSTM machine learning methods and compared them with five other classifiers. It is achieved 91.9% accuracy result for TBED based on Plutchik's emotions in a short text [47].

Table I presents the a list of recent works in further detail.

IV. DATASETS

The dataset collection process is an influential part of every machine learning work, and can affect the quality of the work significantly in both negative and positive

ways. This challenge becomes more important and complicated when dealing with textual datasets because of two reasons. First, manually annotating such datasets in a large-size is a costly and time consuming process. Second, in most cases, assigning labels is related to the humans non-objective cognitive views. Table II presents most of emotion data sets in detail.

V. EMOTION LEXICONS

TBED mechanisms apply emotion lexicons to both lexicon-based or ML-based. Emotion lexicons are lists of english words with their labels. There are two types of emotion lexicons. One is a General Purpose Emotion Lexicon (GPEL) which is from common sources and applies to all the domains and expresses emotion by score such as WordNet-Affect, WPARD, LIWC, NRC, NRC Hashtag [50]. The other emotion lexicon is Domain Specific Emotion Lexicon (DSEL) that is based on specific domains to detect emotion features for labeling [51]. Table III describes the main emotion lexicons.

VI. OPEN ISSUES AND FUTURE WORKS

One of the main challenges of automatic TBED is Word-Sense Disambiguation (WSD) which happens when a word has multiple meanings, and recognizing the right sense of the word is challenging and has remained an open problem in the field of linguistics. One of the important steps of making an ML-based TBED is to fine-tune the existing TBED pipeline by finding out which combination of machine learning techniques and dimension reduction methods results in the best performance of TBED. This step is possible only when adequate volume of labeled datasets are available. Shortage of datasets classified by multiple emotion labels is therefore an obstacle to construct efficient ML-based TBED mechanisms as the annotation process is quite time-consuming and costly. Another challenge is the lack of appropriate emotion lexicon. The currently existing emotion lexicons are not domain-specific and specify the generally comprehensible emotion label of each term; however, emotion labels are strongly context-dependent. This encourages researchers to construct domain-specific lexicons for a given corpus.

VII. CONCLUSION

This paper aimed to provide a survey and multi-disciplinary overview of recent progress on text-based emotion detection. It also studied and classified different emotional psychological models used in different TBED mechanisms. Table IV outlines and compares the performance of the reviewed mechanisms based on year and ratio. We also studied and compared the main methodologies, data sets, and lexicons used in TBED systems. Finally, we reviewed the applications of TBED and addressed open issues in this field.

Name	Year	Author	Size	Type of Data
ISEAR	1997	Scherer	7,666	Sentences
Fairy tales	2005	Alm	15,302	Sentences
SemEval	2007	Strapparava	1,250	News headlines
TEC	2012	Mohammad	21,000	Tweets
CBET	2015	Shahraki	81,163	Tweets
EmoBank	2017	Buechel	10,548	Sentences
CrowdFlower	2016	crowdsourcing	39,740	Tweets
Blogs	2007	Aman	5,205	Sentences
DailyDialogs	2017	Li	13,118	Sentences
Electoral-Tweets	2015	Mohammad	100,000	Tweets
EmoInt	2017	Mohammad	7,097	Tweets
Emotion-stimulus	2015	Ghazi	820, 549	Sentences
FB-valence-arousal	2016	Preot, iuc-Pietro	2,895	Facebook posts
Grounded-Emotions	2017	Liu	2,557	Instances
SSEC	2017	Schuff	4,868	Tweets
Project Gutenberg	2009	Lebert	34,000	Books
Google	2011a	Michel	5.2 million	Digitized books
Hashtag Emotion Corpus	2015	Mohammad	21,051	Tweets

TABLE II: Emotion-related data sets

Name	Author, Year	words	Set of Emotions
Wordnet Affect	Strapparava, 2004	4,787	a hierarchy of emotions
WPARD	Medler, 2005	1,400	positive, negative
LIWC	Pennebaker, 2007	5,000	affective or not, positive, negative, anxiety, anger, sadness
NRC	Mohammad, 2010	14,182	anger, fear, anticipation, trust, surprise, sadness, joy, disgust
NRC hashtag	Mohammad, 2013	32,400	anger, fear, anticipation, trust, surprise, sadness, joy, disgust

TABLE III: Emotion-related lexicons

Author	Year	Measurement Method	Ratio (%)
Strapparava [43]	2008	F_1 score	17.6
Kim [40]	2010	F_1 score	73.3
Cherry [36]	2012	F_1 score	55.2
Wang [41]	2012	F_1 score	65.6
Bandhakavi [38]	2016	F_1 score	60.0
Zad [44]	2020	F_1 score	80.9
Akhtar [46]	2020	accuracy (cosine similarity)	74.8
Chiorrini [45]	2021	F_1 score	89.0
Krommyda [47]	2021	accuracy	91.9

TABLE IV: Comparison of emotion detection ratios of main TBED mechanisms.

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